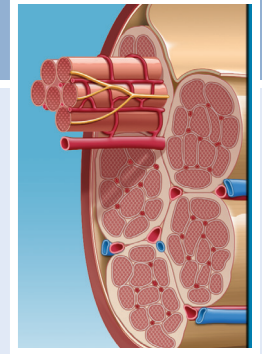


Orthobiologics: Clinical Application of Platelet-Rich Plasma and Stem Cell Therapy

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The human body possesses a tremendous healing potential. However, despite its innate restorative capacity, in many instances the body's ability to heal is limited. Musculoskeletal tissues such as tendon, ligament, and cartilage present challenges to clinicians because these tissues tend to heal slowly because of their limited blood supply and slow cell turnover. Furthermore, conservative management or surgical intervention alone may not reliably recapitulate the normal architecture and function of the injured tissue. Therein lies the potential benefit of biologic therapy, in which the addition of growth factors and reparative cells may not only augment the normal body healing process but also restore normal form and function.

The use of biologic therapy or regenerative medicine has grown exponentially in the field of sports medicine in recent years. Although these emerging therapies may be based on solid preclinical evidence, these therapies are still in the stages of building clinical evidence before they are incorporated into standards of care. This chapter reviews the basic principles and best available clinical evidence on two growing categories of biologic therapy in sports medicine: platelet-rich plasma (PRP) and stem cell therapy.

PLATELET-RICH PLASMA AND SPORTS MEDICINE

The use of human blood concentrates to treat various tendon, ligament, and cartilage disorders and to augment surgical repairs continues to grow within sports medicine. The availability of PRP and its autologous nature allow for easy clinical application without the risk associated with allogenic products. As a result, PRP has been used in several clinical studies for the management of pathologic conditions of ligament, tendon, bone, and cartilage (Fig. 5.1). Although an extensive body of literature on PRP exists, a clear consensus for or against the use of PRP in the treatment of musculoskeletal diseases has not been achieved. What is clear is that not all PRP preparations are created equal,¹⁻⁴ and our understanding regarding the various components of PRP, along with its optimal composition, timing, and delivery, continues to be refined.

Definitions and Properties of Platelet-Rich Plasma

PRP is an autologous concentration of human platelets in a small volume of plasma produced from a patient's own centrifuged

blood. The concentrated platelets contain increased amounts of growth and differentiation factors, which can then be delivered to an injury site to augment the body's natural healing process. The normal human platelet count ranges anywhere from 150,000 to 350,000 per μL . Improvements in bone and soft tissue healing properties have been demonstrated with concentrated platelets of 1,000,000 per μL , and thus it is this concentration of platelets in a 5-mL volume of plasma that has been suggested as one working definition of PRP.^{5,6} A more modern definition of PRP is any plasma fraction that concentrates platelets greater than baseline. A resultant three- to fivefold increase in growth and differentiation factors can be expected with PRP compared with normal nonconcentrated whole blood. PRP preparations are typically further categorized into leukocyte-rich PRP (LR-PRP) preparations, defined as having a neutrophil concentration above baseline, and leukocyte-poor PRP (LP-PRP) preparations, defined as having a leukocyte (neutrophil) concentration below baseline.

Preparation and Composition

Currently more than 16 commercial PRP systems are available on the market, and hence quite a bit of variation exists in the PRP collection and preparation protocol depending on the commercial system being used (Table 5.1). Each commercial system has a different platelet capture efficiency that results in different whole-blood volume requirements to achieve the necessary final platelet concentration for PRP. The commercial systems may also differ in their isolation method (one- or two-step centrifugation), the speed of centrifugation, and the type of collection tube system and operation. In general, whole blood is usually collected and mixed with an anticoagulant factor, such as acid citrate dextrose, sodium citrate, or ethylene diamine tetraacetic acid. Centrifugation is then required to separate red blood cells (RBCs) from platelet-poor plasma (PPP) and the "buffy coat," which contains the concentrated platelets and $\pm$ leukocytes. The RBC and PPP layers may be discarded using various processing techniques to isolate the platelet-concentrated layer (Fig. 5.2). The platelets can then be directly injected into the patient or be "activated" via the addition of either calcium chloride or thrombin, which then causes the platelets to degranulate and release the growth and differentiation factors. Approximately 70% of the stored growth factors are released within the first 10 minutes of activation, and nearly 100% of the growth factors are released

Abstract

Background: Many musculoskeletal tissues have limited healing potential due to their limited blood supply and slow cell turnover. Biologics or regenerative treatments, such as platelet-rich plasma (PRP) and stem cell therapy, have been used with increasing frequency for various tendon, ligament, and cartilage disorders.

Evidence Review: A plethora of proprietary PRP and stem cell preparations are currently commercially available. Although specific composition varies across systems, PRP is typically categorized as leukocyte/neutrophil-rich PRP (LR-PRP) and leukocyte/neutrophil-poor PRP (LP-PRP). The cell source of autologous stem cell preparations is predominantly from bone marrow aspirate and adipose tissue.

Findings: Current evidence suggests that LR-PRP injection is an efficacious treatment for lateral epicondylitis and patellar tendinopathy; and LP-PRP injection is efficacious in treating symptoms of knee osteoarthritis. Although there are fewer clinical studies for stem cell treatments, current evidence supports the use of bone marrow aspirate concentrate (BMAC) as an adjuvant for rotator cuff repair, focal chondral defect repairs, and knee osteoarthritis, with some promising preliminary results for patellar tendinopathy and lateral epicondylitis and as an adjuvant for meniscal repair. There is evidence supporting the use of adipose-derived stem cells in Achilles tendinopathy and knee osteoarthritis and as an adjuvant in focal chondral defect repairs, with promising results in lateral epicondylitis.

Conclusions: Orthobiologic and regenerative medicine treatments, such as PRP and stem cell therapies, have great potential but clinical evidence has yet to demonstrate definitive efficacy. Despite a relative paucity of high-quality data, these technologies have shown promise and may become key modalities in the treatment of musculoskeletal disorders.

Keywords

PRP
platelet-rich plasma
stem cell
adipose
bone marrow
mesenchymal

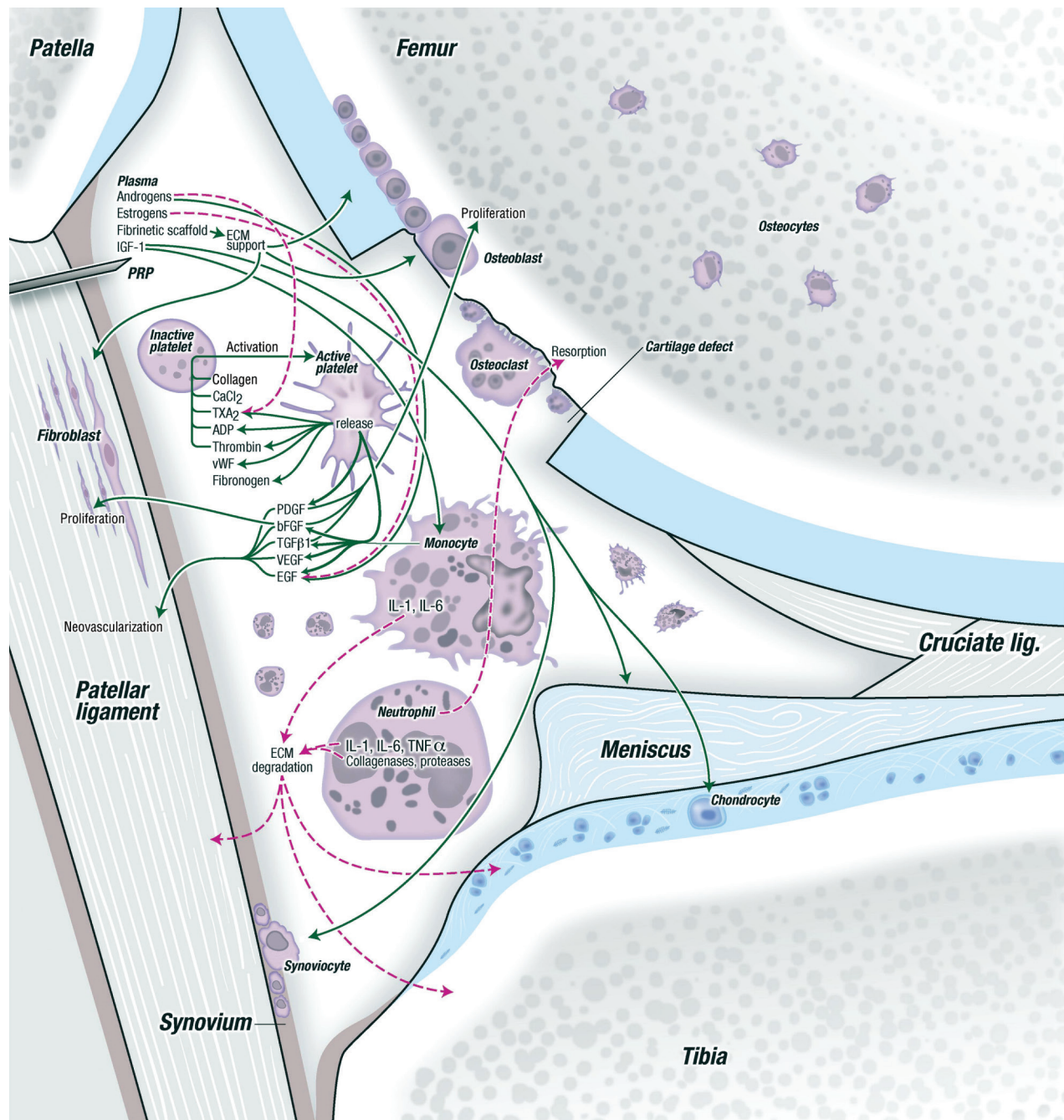


Fig. 5.1 The effect of various components of platelet-rich plasma on different types of tissues surrounding a joint. ADP, Adenosine diphosphate; bFGF, basic fibroblast growth factor; $CaCl_2$, calcium chloride; ECM, extracellular matrix; EGF, epidermal growth factor; IGF, insulin growth factor; IL, interleukin; PDGF, platelet-derived growth factor; PRP, platelet-rich plasma; TGF- β 1, transforming growth factor- β 1; TNF, tumor necrosis factor; TXA_2 , thromboxane A_2 ; VEGF, vascular endothelial growth factor; vWF, von Willebrand factor. (From Boswell SG, Cole BJ, Sundman EA, et al. Platelet-rich plasma: a milieu of bioactive factors. *Arthroscopy*. 2012;28:429–439.)

within 1 hour of activation.^{5,6} Small amounts of growth factors may continue to be produced by the platelet during the remainder of its life span (8 to 10 days).

Platelets contain a milieu of growth factors and mediators in their alpha granules (Table 5.2).²⁵ However, the specific composition of PRP likely varies not only from person to person but also when the isolation process is repeated in the same individual.⁴

Several elements are known to influence the specific makeup of PRP, which includes patient-specific factors and different commercial system preparation methods.^{1,4,7} The variability in the cellular composition of PRP preparations creates challenges in interpretation of the literature regarding the clinical efficacy of PRP.

Our current understanding appears to suggest that PRP with elevated leukocyte content (i.e., LR-PRP) is associated with an

TABLE 5.1 Available Commercial Platelet-Rich Plasma Systems and Characteristics

System	Company	Blood Volume Required (mL)	Concentrated Volume Produced (mL)	Processing Time (min)	PPP Produced?	Gel Activator Available?	Increase in [Platelets] (Times Baseline)	Platelet Capture Efficiency (% Yield)
Leukocyte-Rich PRP								
Angel	Arthrex	52 ¹³⁷	1–20 ^a	17 ¹³⁷	+		10 ^a	56–75% ¹³⁷
GenesisCS	EmCyte	54 ¹³⁷	6 ¹³⁷	10 ¹³⁷	+		4–7 ¹³⁷	61 ± 12% ¹³⁷
GPS III	Biomet	54 ¹³⁷	6 ¹³⁷	15 ¹³⁷	+	+	3–10 ¹³⁷	70 ± 30% ¹³⁷
Magellan	Isto Biologics/ Arteriocyte	52 ¹³⁷	3.5–7 ¹³⁷	17 ¹³⁷	+		3–15 ¹³⁷	86 ± 41% ¹³⁷
SmartPRP 2	Harvest	54 ¹³⁷	7 ¹³⁷	14 ¹³⁷	+		5–9 ¹³⁷	94 ± 12% ¹³⁷
Leukocyte-Poor PRP								
Autologous Conditioned Plasma (ACP)	Arthrex	11 ¹³⁸	4 ¹³⁸	5 ¹³⁸		+	1.3 ¹³⁸	48 ± 7% ¹³⁸
Cascade	MTF	18 ³	7.5 ³	6 ³		+	1.6 ³	68 ± 4% ³
Clear PRP	Harvest	54 ^a	6.5 ^a	18 ^a	+		3–6 ^a	62 ± 5% ^a
Pure PRP	EmCyte	50 ^a	6.5 ^a	8.5 ^a	+		4–7 ^a	76 ± 4% ^a

^aData obtained from manufacturers' promotional literature or internal studies.
PPP, Platelet-poor plasma; PRP, platelet-rich plasma.

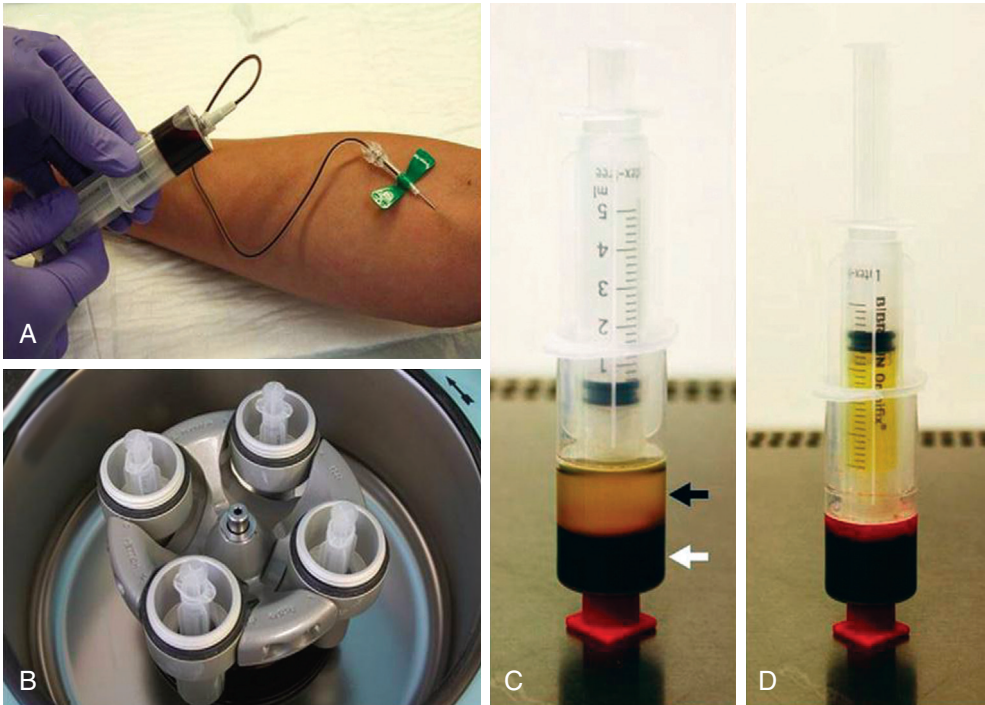


Fig. 5.2 Autologous platelet-rich plasma preparation using a single-step centrifugation system. (A) Autologous whole blood is aspirated in a double-syringe system. (B) The syringe is then placed in corresponding buckets of a desktop centrifuge and spun after balancing. (C) The serum fraction containing the clotting factors, white blood cells, and the platelets (buffy coat) (*black arrow*) is separated from the red blood cell fraction (*white arrow*). (D) By pulling the stamp of the second syringe of the closed system, the serum fraction is extracted from the red blood cell fraction in a sterile manner for further use. (From Steinert AF, Middleton KK, Araujo PH, et al. Platelet-rich plasma in orthopaedic surgery and sports medicine: pearls, pitfalls, and new trends in research. *Oper Tech Orthop.* 2012;22:91–103.)

TABLE 5.2 Platelet-Rich Plasma Growth Factors and Their Physiologic Effects

Factor	Target Cell and Tissue	Function
PD-EGF	Epithelium, endothelial cells Fibroblasts, keratinocytes	Cell growth and recruitment Cell differentiation, skin closure Cytokine secretion
PDGF AB	Fibroblasts, smooth muscle cells, chondrocytes, osteoblasts, mesenchymal stem cells	Potent cell growth, recruitment Blood vessel growth, granulation Growth factor secretion; matrix formation with BMPs (collagen and bone)
TGF- β 1	Endothelial cells, keratinocytes Fibroblasts, lymphocytes, monocytes Osteoblasts	Blood vessel and collagen synthesis Growth inhibition, apoptosis (cell death) Differentiation, activation
IGF-I, II	Bone, blood vessel, skin, other tissues Fibroblasts	Cell growth, differentiation, recruitment Collagen synthesis with PDGF
bFGF	Endothelial cells, smooth muscle, skin Fibroblasts, keratinocytes	Cell growth Cell migration, blood vessel growth
VEGF, ECGF	Blood vessel cells	Cell growth, migration, new blood vessel growth Antiapoptosis

bFGF, Basic fibroblast growth factor; *BMP*, bone morphogenetic protein; *ECGF*, endothelial cell growth factor; *IGF*, insulin-like growth factor; *PD-EGF*, platelet-derived epidermal growth factor; *PDGF*, platelet-derived growth factor; *TGF*, transforming growth factor; *VEGF*, vascular endothelial growth factor.

acute inflammatory response. In an animal model, Dragoo et al.¹ demonstrated this large inflammatory response in vivo with increased cellularity and vascularity in rabbit tendons 5 days after treatment with LR-PRP, compared with LP-PRP. Elevated leukocyte (neutrophil) concentrations present in LR-PRP are also associated with elevated catabolic cytokines, such as interleukin-1 β (IL-1 β), tumor necrosis factor- α (TNF- α), and metalloproteinases,^{7,8} which may antagonize the anabolic cytokines contained within platelets. The clinical ramifications and cellular effects of these different PRP preparations, including leukocyte content, are still currently being elucidated but have begun to allow for more customized PRP treatments to target specific indications.

PLATELET-RICH PLASMA IN TENDON- AND LIGAMENT-RELATED DISORDERS AND REPAIR

Tendon and Ligament Injuries

PRP has been most actively evaluated in the treatment of tendon and ligament injuries. Tendon and ligaments heal through a dynamic process with stages of inflammation, cellular proliferation, and subsequent tissue remodeling. Many of the cytokines found in PRP are involved in the signaling pathways that occur during this restorative process.^{2,5} PRP may also promote neo-vascularization, which may not only increase the blood supply and nutrients needed for cells to regenerate the injured tissue but may also bring new cells and remove debris from damaged tissue. Both these mechanisms of action are particularly attractive in chronic tendinopathy conditions in which the biologic milieu may be unfavorable for tissue healing.

Clinical studies on the use of PRP for tendon and ligament injuries have mainly focused on the areas of the elbow, knee, and ankle. Studies on lateral epicondylitis suggest that LR-PRP may be effective and a reasonable option for patients who have failed to respond to a physical therapy regimen. In a prospective

cohort study, Mishra et al.⁹ evaluated 230 patients who failed to respond to at least 3 months of conservative treatment for lateral epicondylitis. Patients were treated with LR-PRP and followed up for 24 weeks. At 24 weeks, the patients who received LR-PRP reported a 71.5% improvement in their pain scores compared with a 56.1% improvement in the control group ($P = .019$). The percentage of patients reporting significant residual elbow tenderness at 24 weeks was 29.1% in the patient group receiving PRP compared with 54.0% in the control group ($P = .009$). There was a clinically meaningful and significant improvement at 24 weeks in patients treated with LR-PRP versus active control.

In terms of the sustainability of treatment effect, PRP may provide longer continuous relief of symptoms for lateral epicondylitis than do corticosteroids. Peerbooms et al.¹⁰ evaluated the efficacy of PRP versus corticosteroids in 100 patients who had a minimum 6-month history of recalcitrant chronic epicondylitis and had failed to respond to conservative management. Treatment success within this study was defined as, at minimum, a 25% reduction in the visual analog scale (VAS) score or Disabilities of Arm, Shoulder, and Hand score without a repeat intervention after 1 year. Although both groups improved in VAS scores from baseline, 73% (37 of 51 patients) in the PRP group versus 49% (24 of 49 patients) in the corticosteroid group were considered to have a successful response at 1 year ($P < .001$). Furthermore, 73% (37 of 51 patients) in the PRP group versus 51% (25 of 49 patients) in the corticosteroid group noted improved Disability of Arm, Shoulder, and Hand scores at 1 year ($P = .005$). Patients who received PRP also continued to report symptom relief 1 year after receiving the injection. In contrast, the short-term benefits of corticosteroids began to wane after 12 weeks. In a separate report, the improvement within this group of patients who received PRP continued to be noted 2 years after the PRP injection.¹¹

In addition to the use of PRP to treat lateral epicondylitis, results from randomized controlled trials support the use

of LR-PRP to treat chronic refractory patellar tendinopathy. Dragoo et al.¹² evaluated 23 patients with patellar tendinopathy on examination and magnetic resonance imaging (MRI) who had failed conservative management. Patients were randomized to receive ultrasound-guided dry needling alone or with injection of LR-PRP. Patients were followed for greater than 26 weeks. At 12 weeks, the PRP group had improved, as measured by Victorian Institute of Sport Assessment-Patella (VISA-P) score, significantly more than the dry needling group ($P = .02$). However, the difference was not significant at greater than 26 weeks ($P = .66$), suggesting that the benefit of PRP for patellar tendinopathy may be earlier improvement of symptoms. Vetrano et al.¹³ also reported the benefit of PRP injections for treatment of chronic refractory patellar tendinopathy. Forty-six patients with ultrasound-confirmed chronic unilateral patellar tendinopathy were randomized to received two PRP injections over 2 weeks or 3 sessions of focused extracorporeal shock wave therapy (ECSWT). Although there was no significant difference between groups at 2-month follow-up, the PRP group showed statistically significant improvement, as measured by VISA-P and VAS, over ECSWT at 6-month and 12-month follow-up, and as measured by Blazina scale score at 12-month follow-up ($P < .05$ for all).

Compared with findings for lateral epicondylitis and patellar tendinopathy, the higher-level data with regard to the use of PRP in patients with Achilles tendinopathy have been less promising. In a prospective randomized trial, de Vos et al.^{14,15} found no significant benefits with PRP versus a saline solution injection as an adjunct to eccentric exercises for mid-Achilles tendinosis. The authors reported no significant differences in Achilles tendon structure, the degree of neovascularization, and clinical outcome compared with the saline solution group. In a follow-up study on the same patients, de Jonge et al.¹⁶ similarly reported no significant benefit in terms of pain reduction, activity level, and tendon appearance on ultrasound at 1 year after injection of PRP for chronic Achilles tendinopathy.

Rotator Cuff Repair

Most studies on the use of PRP products as an adjuvant to tendon repairs have focused on rotator cuff repairs. Some studies used a fibrin matrix as a carrier for the PRP,^{17,18} whereas others injected PRP directly into the repair site.^{19–21} In a randomized controlled study, Castricini et al.¹⁷ reported outcomes based on Constant score and MRI-evaluated integrity of the repaired rotator cuff in 88 patients after 16 months follow-up. There was no significant benefit with PRP for small- (<1 cm) and medium-sized (1 to 3 cm) rotator cuff tear repairs. The authors found no significant differences between the PRP and the control group in terms of the Constant score, tendon thickness, and repeat rupture rate.¹⁷ In another randomized controlled trial, Weber et al.¹⁸ similarly reported no significant improvement using PRP in perioperative morbidity, clinical outcomes, or structural integrity. The study randomized 60 patients and found that structural results correlated with age and size of tear but did not differ between the PRP-treated group and control.

Randelli et al.¹⁹ evaluated use of PRP in a randomized control study of 53 single-tendon rotator cuff repairs. Patients in the

PRP-treated group reported short-term pain reduction and increased Simple Shoulder Test scores at early postoperative periods, with all clinical scores superior for PRP at 3 months after surgery ($P < .05$). In their subgroup analysis, patients who had undergone repairs of small-sized tears (grade 1 and 2 tears, with less retraction) had lower repeat tear rates (9 of 16 patients or 40% vs. 12 of 19 patients or 52%) and increased external rotation strength throughout the 24-month follow-up period ($P < .05$). Jo et al.²⁰ assessed the efficacy of PRP augmentation in a randomized controlled trial of 74 patients undergoing arthroscopic repair of medium to large rotator cuff tears. There was no difference between PRP and control in the Constant score at 3 months nor in VAS for pain, strength, or functional scores. However, the retear rate of the PRP group (3.0%) was significantly lower than the control group (20.0%) ($P = .032$) and the PRP group had a lower decrease in the cross-sectional area (CSA) of the supraspinatus muscle ($P = .014$), indicating an improved structural outcome. Finally, Pandey et al.²¹ used an only moderately concentrated LP-PRP preparation in a randomized study of 102 patients undergoing repair of medium and large degenerative tears. VAS scores were significantly lower in the PRP group at 1, 3, and 6 months ($P = .0005$) but not later. Constant scores were significantly better in the PRP group at 12 and 24 months ($P < .01$) and UCLA scores were significantly higher at 6 and 12 months ($P = .002$). At 24 months, the retear rate in the PRP group was significantly lower than in the control group ($P = .01$), with subgroup analysis showing significance only for large tears ($P = .03$).

Achilles Tendon Repair

Data are limited on the use of PRP in the repair of acute Achilles tendon tears, and findings in the existing literature are conflicting. Schepull et al.²² evaluated the use of PRP in Achilles tendon repair in a randomized study ($n = 30$). Although no differences were reported with regard to tendon elasticity and heel raise index between the PRP and control groups, the authors did note a lower Achilles Tendon Total Rupture score among the PRP group, which suggests a detrimental effect of PRP on subjective outcome after repair. However, Zou et al.²³ enrolled 36 patients with acute Achilles tendon rupture in a prospective randomized controlled study using intraoperative LR-PRP injection versus repair without PRP. Patients were followed for 24 months. Patients from the PRP group had better isokinetic muscle at 3 months and had higher SF-36 and Leppilahti scores at 6 and 12 months, respectively ($P < .05$ for all). Ankle range of motion (ROM) was also significantly better in the PRP group at all time points of 6-, 12-, and 24-months ($P < .001$).

Anterior Cruciate Ligament Surgery

The success of anterior cruciate ligament (ACL) surgery not only hinges on technical factors (e.g., graft tunnel placement and graft fixation) but also biologic healing of the ACL graft. Given its potential for improving tissue vascularity and ligament healing, PRP has been used to augment ACL graft maturation and graft–bone tunnel incorporation after reconstruction. Within the research literature, ACL graft maturation tends to be assessed with MRI. The assumption is that a low homogenous intensity

signal on T2- and proton density-weighted MRI is likely indicative of a healthy maturing ACL graft. In terms of the effect of PRP on ACL graft maturation, some studies have demonstrated improved graft maturation with PRP,^{24–27} whereas others reported a lack of significant differences.^{28,29} Authors of a recent systematic review of 11 controlled trials, which included studies in which statistical significance was not reached, concluded that PRP likely improves ACL graft maturation, by up to 50%. The authors pointed to insufficient sample size as a potential rationale for lack of statistical significance despite MRI improvement in some metrics measuring ACL graft maturation.³⁰

The other component to successful biologic healing of an ACL graft is graft–bone tunnel incorporation. The existing literature on the use of PRP to augment healing of the graft–bone interface is inconclusive at best. Vogrin et al.²⁹ evaluated the effects of PRP gel treatment for hamstring autograft ACL reconstruction in a controlled double-blinded study. MRI was used after the operation to assess vascularization along the ACL graft–bone interface. The authors reported evidence of improved vascularization along the interface at 3 months with use of PRP, but the observed benefit dissipated by 6 months after the surgical procedure. Other studies have similarly reported limited to no evidence to support the use of PRP to augment ACL graft–bone tunnel incorporation.^{24,31} Of note is that nearly all of the studies used an LR-PRP formulation and LR-PRP formulations increase local tissue inflammation, which may delay or alter healing.¹

One final point of consideration is whether any of the observed benefit of PRP on ACL graft maturation or graft–tunnel healing would translate into improved clinical results. The current best available evidence seems to suggest no significant benefit for functional outcome with use of PRP.^{24,28,32} Ventura et al.³² found no differences in Knee Injury and Osteoarthritis Outcome score (83 vs. 84 points), KT-1000 (0.8 mm vs. 1.2 mm), or Tegner scores (0.9 vs. 0.8 difference preoperative to postoperative) between the PRP-treated group and control subjects at 6 months despite reporting a significant difference in graft appearance. Orrego et al.²⁴ similarly noted no significant benefit in both Lysholm and International Knee Documentation Committee scores at 6 months after the operation despite identifying a positive effect of PRP on graft maturation. In summary, evidence from the current literature suggests that PRP may improve the rate at which ACL grafts achieve a low signal on MRI T2-weighted imaging but have little to no effect on graft–tunnel incorporation. A demonstrable benefit in patient outcome after use of PRP in patients undergoing ACL surgery is also lacking.

PLATELET-RICH PLASMA IN CARTILAGE RESTORATION

When considering biologic approaches to cartilage problems, it is important to understand that focal cartilage injuries differ from arthritis in terms of joint biology, homeostasis, and levels of metalloproteases and inflammatory cytokines.³³ Therefore PRP applications and clinical results for cartilage lesions and osteoarthritis may be quite different for either condition. The idea of using PRP for cartilage repair is based on in vitro basic science literature that growth factors released by the platelet

alpha granules may increase the synthetic capacity of chondrocytes through upregulation of gene expression, proteoglycan production, and deposition of type II collagen.^{34–36}

Although very few clinical reports or trials have evaluated the use of PRP for treatment of focal cartilage lesions, Dhollander et al.³⁷ reported a pilot case series of five patients surgically treated with autologous matrix-induced autologous chondrocyte implantation (MACI) combined with a PRP gel. The authors reported that although VAS scores for pain decreased, the lesion was not filled after 12 and 24 months based on MRI evaluations, and intralesional osteophytes developed in three of the five patients.

Most clinical reports on the use of PRP for cartilage injury involve patients with osteoarthritis of the knee. A recent meta-analysis by Riboh et al.³⁸ compared LP-PRP and LR-PRP in the treatment of knee osteoarthritis and found that LP-PRP injections resulted in significantly improved WOMAC scores compared with hyaluronic acid (HA)^{39,40} or placebo.^{41,42} Patel et al.⁴¹ performed a prospective randomized trial comparing single- or double-injection LP-PRP with saline solution in 78 patients with early osteoarthritis. They concluded that a single injection of PRP was as effective as a double injection. On the other hand, Filardo et al.⁴³ enrolled 192 patients in a randomized controlled study and found no difference between LR-PRP and HA, providing further evidence that LP-PRP may be an effective choice for treatment of osteoarthritis symptoms and LR-PRP appears not to be.³⁸ The biological basis for this may be in the relative level of inflammatory versus antiinflammatory mediators present in LR-PRP and LP-PRP. Inflammatory mediators TNF- α , IL-6, interferon- γ (IFN- γ), and IL-1 β are increased significantly in the presence of LR-PRP,^{7,44,45} whereas injection of LP-PRP increases IL-4 and IL-10, which are antiinflammatory mediators. IL-10 specifically was found to be helpful in the treatment of hip osteoarthritis⁴⁶ and may also suppress the release of the inflammatory mediators TNF- α , IL-6, and IL-1 β , and block the inflammatory pathway by neutralizing nuclear factor- κ B activity.^{7,44,46} In addition to its deleterious effects on chondrocytes, LR-PRP may also fail to help treat osteoarthritis symptoms due to its effect on synoviocytes. Braun et al.⁴⁷ found that treatment of synovial cells with LR-PRP or erythrocytes resulted in significant proinflammatory mediator production and cell death.

PLATELET-RICH PLASMA IN MENISCAL REPAIR

The idea of augmenting meniscus repair with growth factors is not a recent one. In 1988 Arnoczky et al.⁴⁸ proposed the use of an exogenous fibrin clot to stimulate a reparative response in the avascular portion of the meniscus. In 1990 Henning et al.⁴⁹ reported that the failure rate for an isolated meniscus tear repair was 41% without a fibrin clot versus 8% with a fibrin clot. Bhargava et al.⁵⁰ demonstrated that platelet-derived growth factor increases the number of cells and tissue formation in a meniscus defect explant model. Several other in vitro and in vivo studies have demonstrated the potential for cytokines found in PRP to improve meniscal cell growth and meniscus repair healing.^{51–55} Ishida et al.⁵³ performed a combined in vivo and in vitro study demonstrating that PRP led to improved meniscal repair in

full-thickness 1.5-mm diameter defects in the avascular zone of the rabbit meniscus.

Augmenting meniscal repair with PRP is logical because it may deliver growth factors to the healing tissue, but a paucity of human clinical data exists. Pujol et al.⁵⁶ evaluated 34 young patients (age 13 to 40, median age 28) with symptomatic grade 2 or 3 horizontal meniscus tears in a case-control study. PRP was injected prior to skin closure in an open meniscal repair versus open meniscal repair alone. One year after surgery, KOOS scores for pain, symptoms, daily activities, sports, and quality of life were superior in the PRP-treated group, reaching statistical significance for pain and sports parameters ($P < .05$). Five out of the 17 cases in the PRP group had complete disappearance of any abnormal signal within the repaired meniscus on MRI, compared with none in the open repair alone group ($P < .01$).

PLATELET-RICH PLASMA FOR MUSCLE INJURIES

The use of PRP in the treatment of muscle injuries has attracted a significant amount of interest in recent years. Similar to tendon healing, the steps in muscle healing involve the initial inflammatory response, which is then followed by cell proliferation, differentiation, and tissue remodeling. Hamid et al.⁵⁷ conducted a single-blind randomized study of 28 patients with grade 2 hamstring muscle injuries comparing an injection of LR-PRP with a rehabilitation program versus rehabilitation alone. The group treated with LR-PRP was able to return to play (RTP) in a significantly shorter amount of time compared with control (mean time in days, 26.7 vs. 42.5, $P = .02$), but structural improvement was not achieved. In a double-blind randomized controlled trial, Reurink et al.⁵⁸ evaluated 80 patients comparing PRP injections to placebo saline injections, with all patients receiving standard rehabilitation. The patients were followed for 6 months and there were no significant differences in RTP time or with reinjury rate.

Although clinical studies have not found PRP to be efficacious in treating muscle injuries, advances in laboratory research may lead to improved understanding of treatment modalities. In vitro studies have found that PRP is capable of leading to myoblast proliferation but not to myoblast differentiation,⁵⁹ a requisite step in producing muscle tissue. Furthermore, growth factors contained in platelets, specifically myostatin (MSTN) and transforming growth factor- β 1 (TGF- β 1), are actually detrimental to muscle regeneration.^{60,61} Miroshnychenko et al.⁶² found in vitro that treatment with PPP or PRP with a second centrifugation step to remove the platelets induced myoblasts into muscle differentiation. This suggests that perhaps the most beneficial treatment of muscle injuries may be with PPP, although in vivo animal studies followed by human clinical trials will be necessary to further explore this treatment option in the future.

STEM-CELL-BASED THERAPY IN SPORTS MEDICINE

A fundamental limitation in healing of musculoskeletal tissue is the relative avascular nature of certain tissues, such as cartilage

and tendon, which poses a barrier to mobilization of reparative cells and mediators to the site of injury or damage. Proinflammatory cytokines and mitogenic factors are sometimes unable to recruit progenitor cells in sufficient numbers to meet the regenerative needs of the tissue. In a way, this remains an inadequacy of PRP therapy in that the injection of autologous cytokines and growth factors may not achieve its full beneficial effect in stem cell-poor tissues, which may be a contributing factor in the mixed efficacy of PRP and cellular therapy, depending upon the clinical indication. The advent of cell-based therapies has been seen as an intriguing advancement in meeting this deficiency in musculoskeletal tissue healing.⁶³

Cell-based approaches may use cells that have already differentiated into a specific tissue type and function (such as autologous chondrocyte implantation [ACI] for cartilage repair) or undifferentiated multipotent stem cells, mesenchymal stem cells (MSCs),⁶⁴ that retain the capacity to adapt their form and function based on environmental cues. MSC populations were initially discovered in bone marrow but have been isolated from other mesenchymal tissues, including adipose tissue, skin, synovial fluid, periosteum, umbilical cord blood, placenta, and amniotic fluid.^{65–67} MSCs are now known to originate as pericytes^{68,69} and can differentiate into bone, cartilage, tendon, muscle, and adipose tissues.^{70,71} Although MSCs can regenerate various mesenchymal tissues in vitro and some degree of direct engraftment into the repair tissue is suspected in vivo, the greatest therapeutic impact of MSCs in vivo are their trophic and immune-modulatory effects via paracrine mechanisms (Fig. 5.3).^{67,72–74} MSCs secrete growth factors and chemokines to promote cell proliferation (such as TGF- α , TGF- β , hepatocyte growth factor [HGF], endothelial growth factor [EGF], fibroblast growth factor [FGF]-2, and insulin-like growth factor [IGF]-1)^{75,76} and angiogenesis (vascular endothelial growth factor [VEGF], IGF-1, EGF, and angiopoietin-1),⁷⁷ as well as possessing antiapoptotic potential.^{78–80} MSCs may also play a potent role in regenerative processes through their antiinflammatory effect through secretion of prostaglandin 2, HGF, stromal cell-derived factor 1 (SDF-1), nitrous oxide, indoleamine 2,3-dioxygenase (IDO), IL-4, IL-6, IL-10, IL-1 receptor antagonist, and soluble TNF- α receptor.^{81,82} as well as through inhibition of inflammatory cells such as T cells, natural killer (NK) cells, B cells, and other immune cells.⁸³ Furthermore, it is believed that MSCs also possess antimicrobial properties through upregulation of IDO and secretion of LL-37 (a member of the cathelicidin peptide family)⁸⁴ in response to microbes.^{85,86}

Definitions and Preparations of Stem Cell Therapy

In 2006 the International Society for Cellular Therapy established a minimum set of criteria for defining MSCs. According to their criteria, MSCs: (1) must be plastic adherent; (2) must express cell surface antigens CD105, CD73, and CD90; (3) must not express cell surface antigens CD45, CD34, CD14, CD11b, CD79 α , CD19, or HLA-DR; and (4) must be capable of trilineage induction (osteoblasts, adipocytes, and chondroblasts) in vitro.⁸⁷ However, these criteria are based on characteristics of the cultured cells in vitro and may not reflect the native phenotype in vivo.^{88,89} For this reason, among others, the criteria and strict definition of an MSC continue to evolve. Assays for MSC populations

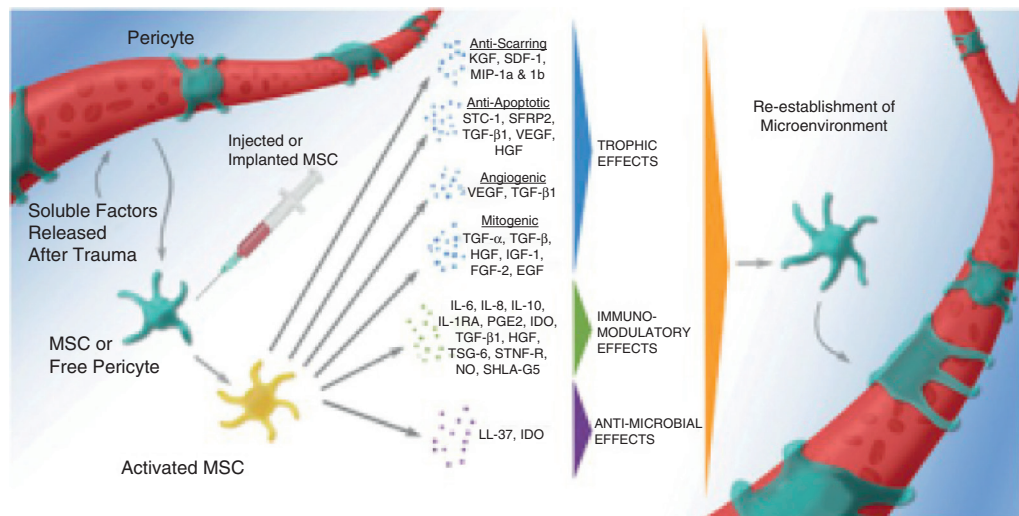


Fig. 5.3 Pericytes are stimulated by soluble growth factors and chemokines to become activated mesenchymal stem cells (MSCs), which respond to the microenvironment by secreting trophic (mitogenic, angiogenic, antiapoptotic, or scar reduction), immunomodulatory, or antimicrobial factors. (From Murphy MB, Moncivais K, Caplan AI. Mesenchymal stem cells: environmentally responsive therapeutics for regenerative medicine. *Exp Mol Med*. 2013;45[11]:e54.)

measure their colony-forming capacity, or colony-forming unit (CFU), which is the number of single cells able to give rise in vitro to colonies of progeny cells.⁹⁰ Although the optimal method of autologous MSC preparation or isolation has yet to be determined,⁸⁹ current practices typically involve harvesting MSC-containing tissue from a source area of high yield and then concentrating the tissue to further isolate the MSCs before delivering them to the area of treatment. Stem cell therapy with MSCs for sports medicine indications most commonly uses MSCs isolated from bone marrow aspiration, or bone marrow aspirate concentrate (BMAC), or from adipose tissue, otherwise known as adipose-derived stem cells (ADSCs).⁶³

In a BMAC preparation, bone marrow is percutaneously aspirated from a marrow harvest site and mixed with an anti-coagulant such as heparin, then concentrated using centrifugation.⁹¹ Several different marrow harvest sites have been used, such as the iliac crest, proximal tibia, proximal humerus, and calcaneus. Analyses have identified the iliac crest as the marrow harvest site with the greatest MSC yield,⁹² with posterior iliac crest preferred over anterior iliac crest due to its superior safety profile (Fig. 5.4).⁹³ Various advances have sought to optimize aspiration technique and improve MSC yield. Muschler et al.⁹⁴ first looked at the effect of aspiration volume on the resulting MSC concentration. Their findings demonstrated that MSC concentration was maximized with aspirations of 2 mL and that aspirations larger than 4 mL had peripheral blood dilution.⁹⁴ Hernigou et al.⁹¹ further examined technique using different sized syringes for bone marrow aspiration and determined that for a given marrow harvest volume; using 10 mL syringes resulted in a 300% higher MSC concentration than using 50 mL syringes. Further studies have suggested larger-volume aspirations are best performed from multiple harvest sites, through different harvest locations or through advancement of the aspiration needle deeper below the cortex, instead of from a single location⁹⁵; and rapid



Fig. 5.4 Bone marrow aspirate is extracted from the iliac crest prior to concentration. (From Murawski CD, Duke GL, Deyer TW, Kennedy JG. Bone marrow aspirate concentrate [BMAC] as a biological adjunct to osteochondral lesions of the talus. *Tech Foot Ankle Surg*. 2011;10[1]:18–27.)

aspiration of the syringe increases MSC yield but increases patient discomfort.⁹⁶ Despite improved harvest techniques, MSCs still represent only a small fraction of the cells obtained from BMAC; of the 6–18 × 10⁶ nucleated cells per mL of bone marrow aspirate obtained, only 0.001% to 0.01% are MSCs, whereas the remainder are various myelopoietic cells, erythropoietic cells, or peripheral blood cells, including platelets.^{70,97}

In an ADSC preparation, adipose tissue is obtained through either a lipoaspiration technique or an operative tissue resection (open or arthroscopic).^{98,99} The lipoaspiration harvesting technique typically involves percutaneous infiltration of subcutaneous

fat (e.g., abdominal or gluteal) with a tumescent fluid, most often a mixture of local anesthetic and saline.¹⁰⁰ After the tumescent fluid is in the subcutaneous tissue and the area sufficiently anesthetized, the adipose tissue is aspirated using manual or mechanical suction.⁹⁹ The arthroscopic technique of harvesting adipose tissue from the infrapatellar fat pad was described by Dragoo et al.¹⁰¹ [AJSM paper, Dragoo], whereby the fat pad is resected using a motorized shaver system which allows the adipose tissue to be collected in a sterile container. The mononuclear fraction of the adipose tissue, also known as the stromal vascular fraction (SVF), contains the MSCs, as well as endothelial precursor cells (EPCs), monocytes, and other vascular and immune cells.¹⁰² Once the adipose tissue is obtained, whether by aspiration or arthroscopic resection, the tissue is usually mechanically fractionated, using either syringes using decreasing diameter Luer-Lok adapters or with proprietary systems, and then centrifuged in order to isolate the SVF from the majority of the adipocytes (Fig. 5.5).¹⁰³ Although adipose tissue contains fewer nucleated cells per milliliter of tissue ($0.6\text{--}2.0 \times 10^6$ per mL of lipoaspirate),¹⁰⁴ MSCs make up a greater proportion of these cells (1% to 4%),¹⁰⁵ therefore making MSCs 100 to 1000 times more frequent in adipose tissue compared with bone marrow aspirate on a per volume basis.^{105,106} However, while the presence of platelets and peripheral blood cells makes supplementation with PRP unnecessary for BMAC, ADSC on its own may lack some required growth factors. Therefore optimal treatment with ADSC may involve the addition of PRP to augment its regenerative potential.¹⁰⁷

Challenges for Application of Cell-Based Therapies

Efficient isolation and delivery of MSCs remains one of the challenges in using stem-cell-based therapies as a point-of-care treatment in clinical scenarios. Hernigou et al.¹⁰⁸ suggested that

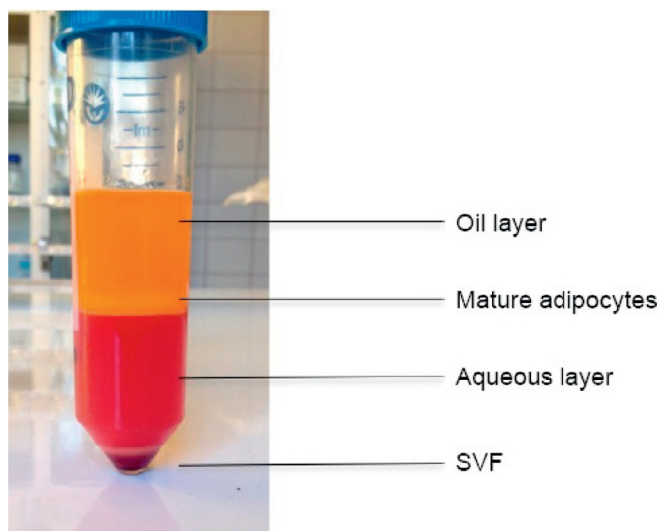


Fig. 5.5 Adipose-derived stem cell preparation with labels denoting the oil layer, mature adipocyte layer, aqueous layer, and mononuclear stromal vascular fraction layer (SVF). (From Malec K, Goralska J, Hubalewska-Mazgaj M, et al. Effects of nanoporous anodic titanium oxide on human adipose derived stem cells. *Int J Nanomedicine*. 2016;11:5349–5360.)

the success of bone marrow aspirate grafting appears to be related to the number of progenitor cells in the aspirate. Several centrifugation systems are now commercially available to concentrate MSCs, but the question regarding the optimal number of cells needed to achieve healing in different musculoskeletal tissues remains. A significant obstacle for the optimization of cell therapy is related to federal regulations, particularly in the United States, governing use of autologous stem cells, including MSCs. Currently no method has been approved by the US Food and Drug Administration (FDA) to expand or manipulate MSCs, such as culture expansion or enzymatic digestion, which limits one's ability to optimize cell concentration and characteristics prior to implantation. The FDA regulates cell therapies primarily under sections 351 and 361, which are subject to different levels of regulation.¹⁰⁹ Products registered under section 351 require an investigational new drug exemption and biologics license application on file with the FDA, whereas products regulated under section 361 bypass the bulk of the requirements. To be eligible for regulation under section 361, cell preparations must be minimally manipulated, only used for homologous use, must not be combined with a drug or device, and must be used autologously.¹¹⁰ For this reason, controversy exists around the potential use of allogeneic umbilical cord blood or amniotic sources of MSC therapy because currently no such product has been registered with the FDA under section 351.¹¹¹

CELL-BASED THERAPY IN TENDON AND LIGAMENT HEALING AND REPAIR

Tendons and ligaments are relatively hypocellular and hypovascular connective tissues,¹¹² and this relative lack of cellular machinery and poor vascularity results in poor healing potential after acute or overuse injuries.⁶⁷ These tissues often heal by reactive scar formation rather than true regeneration, leading to altered function.¹¹³ In response to this problem, biologic augmentation with use of cell-based therapy has been investigated. MSC therapy is thought to improve tendon and ligament healing through its immunomodulatory effects by reprogramming macrophages from a proinflammatory M1 phenotype to an anti-inflammatory M2 phenotype.^{114,115} Saether et al.¹¹⁶ evaluated MSC therapy for enhancing MCL healing in rats, finding that treatment with a low dose of MSCs (1×10^6 cells) resulted in lower amounts of inflammatory cytokines (IL-1 α , IL-12, and IFN- γ) and significant improvements in procollagen I, proliferating cells, and endothelialization. Mechanical testing also found improved strength and stiffness, thus demonstrating improved functional healing. Nixon et al.¹¹⁷ evaluated the reparative properties of adipose-derived MSCs in an equine collagenase-induced tendon injury model. Compared with tendons treated with saline solution, the tendons injected with stem cells demonstrated decreased inflammation and significantly improved collagen fiber architecture and organization 6 weeks after MSC treatment.

Although clinical studies are limited, early data are promising for cell-based approaches to ligament and tendon disorders. Pascual-Garrido et al.¹¹⁸ evaluated the use of autologous MSCs from BMAC for chronic patellar tendinopathy in eight patients. The authors were able to demonstrate both structural and

symptom improvements in this small sample size. Although most of the gains were made in the first 12 months, the authors reported sustained benefit after injection of MSCs as long as 5 years. In a randomized controlled study, de Girolamo et al.¹¹⁹ compared injection of ADSCs against injection of PRP in 56 patients with Achilles tendinopathy recalcitrant to conservative treatment. Both groups had significant improvements in VAS pain, VISA-A, and AOFAS scores ($P < .05$), with the ADSC group improving faster but with equivalent results between groups after 6 months. Lee et al.¹²⁰ examined the use of ADSCs from an allogeneic source in the treatment of lateral epicondylitis in 12 patients followed over 52 weeks. Throughout the entire follow-up period, all patients had improved VAS pain scores and elbow function scores, along with significant decrease in tendon defect areas as measured on ultrasound.

Clinical data are also limited with regard to use of MSCs for tendon repairs, which revolve around rotator cuff repairs, but have shown some encouraging preliminary results. Ellera Gomes et al.¹²¹ evaluated 14 patients with complete rotator cuff tears (one to three tendon tears) who underwent a rotator cuff repair along with a single injection of BMAC. MRI analysis 12 months after surgery revealed good tendon integrity in all the subjects, with favorable clinical outcomes, although no control group was included in the study. In a case-control study, Hernigou et al.¹²² followed 90 patients for 10 years after rotator cuff repair, with 45 patients receiving BMAC injection intraoperatively and 45 patients receiving single-row surgery alone. At 6-month follow-up, 100% of the patients receiving BMAC had healed versus 67% of the control group. At 10-year follow-up, 87% of the patients that received BMAC had intact tendons compared with only 44% of the control group. Among the patients receiving BMAC, the retear rate was inversely correlated to the concentration of MSCs delivered because patients with loss of tendon integrity received fewer MSCs compared with those that stayed intact. Although identifying a convenient localized source of MSCs during shoulder surgery could allow for easier augmentation of rotator cuff repairs, patients with symptomatic rotator cuff tears have decreased MSCs counts (30% to 70% less) at the greater tuberosity of the proximal humerus compared with those without rotator cuff injury.¹²³

CELL-BASED THERAPY IN CARTILAGE REPAIR

Although no therapy or procedure has been shown to stop degenerative joint destruction in osteoarthritis, a significant cause of population-level morbidity, it is hoped that cell-based therapies can slow down or reverse articular cartilage damage through regenerative mechanisms. MSCs for chondrocyte differentiation can be derived from synovium, bone marrow, adipose, and muscle.¹²⁴ Stem cells for the treatment of osteoarthritis are primarily delivered via intra-articular injections, with or without accompanying surgical intervention.

Surgical delivery of MSCs with a scaffold is typically performed for focal articular cartilage lesions. Nejadnik et al.¹²⁵ reported their results comparing a cohort of 36 patients treated with ACI and a cohort of 36 patients treated with culture-expanded bone marrow-derived MSCs. The authors reported no statistically

significant difference between the two treatments in functional evaluation. The authors also obtained biopsy specimens of seven patients (four in the MSC group and three in the ACI group) during second-look arthroscopy, which demonstrated hyaline-like cartilage in both.¹²⁵ Gobbi et al.¹²⁶ performed a comparative study on 37 patients with patellofemoral chondral lesions. Nineteen patients were treated with MACI and 18 with nonexpanded BMAC implantation, all with hyaluronan scaffold. After 3-year follow-up, both groups showed significant improvement ($P = 0.001$) with no significant difference between groups; 76% of MACI-treated patients showed complete filling of their lesion compared with 82% of BMAC-treated patients.¹²⁶ In a randomized controlled trial, Koh et al.¹²⁷ assessed the addition of ADSC injection with microfracture surgery versus microfracture surgery alone in 80 patients with femoral cartilage defect greater than 3 cm². The patients were followed for 24 months clinically and with MRI. The group treated with ADSC had significant improvement in KOOS pain score compared with control ($P = .034$). On MRI, 65% of ADSC-treated patients had complete cartilage coverage compared with 45% of control patients and significantly better signal intensity with 80% of ADSC-treated patients having normal or near-normal signal intensity compared with 72.5% of control.¹²⁷

Clinical trials for intra-articular injections of MSCs for treating osteoarthritis have also been promising. Wong et al.¹²⁸ evaluated the use of intra-articular injections of culture-expanded BMAC in a randomized trial. Fifty-six patients with unicompartmental knee osteoarthritis and genu varum all underwent microfracture and medial opening-wedge high tibial osteotomy (HTO) surgery and were then randomized to receive an injection of cultured BMAC with hyaluronic acid versus hyaluronic acid alone 3 weeks after surgery. Although both treatment arms demonstrated clinical improvement, BMAC-treated patients had significantly better Tegner ($P = .021$), Lysholm ($P = .016$), and IKDC scores ($P = .001$) at 1-year postoperative. The BMAC group also had significant improvement by MRI evaluation ($P < .001$).¹²⁸ Koh et al.¹²⁹ looked at the intra-articular injection of ADSC plus PRP versus PRP alone in a randomized trial of patients undergoing HTO for genu varum. The group receiving ADSC injection with their treatment had significantly greater improvements in VAS pain ($P < .001$), KOOS pain ($P < .001$), and KOOS symptoms ($P = .006$) scores. Furthermore, ADSC-treated patients had significantly better cartilage healing ($P = .023$) on second-look arthroscopy.¹²⁹ Kim et al.¹³⁰ evaluated 41 patients (75 knees) with Kellgren-Lawrence grade 1 to 4 osteoarthritis of the knee. They underwent a single BMAC injection with adipose tissue scaffold and were followed for 1 year. At 12 months, there was significant improvement in clinical pain and function scores (as measured by VAS, KOOS, SF-36, IKDC, and Lysholm) compared with baseline and furthermore demonstrated that patients with more advanced Kellgren-Lawrence (K-L) grade had significantly poorer results ($P = .002$). In a randomized controlled study, Vega et al.¹³¹ compared intra-articular injection of culture-expanded allogenic BMAC to hyaluronic acid in 30 patients with K-L grade II-IV knee osteoarthritis. The patients were followed for 12 months, and patients treated with MSCs had significant improvement in clinical pain scores and MRI cartilage improvement both in quantity and quality of articular cartilage. Michalek

et al.¹³² evaluated the use of ADSC with PRP in 1114 patients with knee or hip osteoarthritis in a multicenter case-controlled study. The patients were followed up for a mean of 17.2 months (range, 2.1 to 54.3 months) and measured modified KOOS/HOOS scores. At 12 months, 91% of patients treated with ADSC with PRP reported at least 50% improvement in symptoms, whereas less than 1% of patients were nonresponders, with four patients undergoing arthroplasty. In subgroup analysis, there was no difference in groups where the ADSC was treated with collagenase or not.

CELL-BASED THERAPY IN MENISCAL REPAIR

The potential of stimulating meniscus repair with use of bone marrow–derived cells has been described by Freedman et al.¹³³ by performing a microfracture adjacent to the PCL femoral origin to allow release of bone marrow elements into the joint. Meniscal repair performed with concomitant ACL reconstruction has historically demonstrated better healing rates, potentially as a result of the bone marrow elements in the joint due to drilling of the bone tunnels.⁴⁹ Several animal models have demonstrated the effectiveness of MSC augmentation in meniscal healing.^{134,135} Although there is a great amount of available data from preclinical studies, the available evidence for use of stem cells in humans to augment meniscal repairs is much smaller. Vangsness et al.¹³⁶ performed the first randomized controlled trial assessing intra-articular injection of MSCs following partial meniscectomy. In this trial, 55 patients underwent a partial medial meniscectomy and were then randomized to receive low-dose allogenic MSCs, high-dose allogeneic MSCs, or hyaluronic acid (control). After 1-year follow-up, a significant increase in meniscal volume (defined as a 15% threshold) on MRI in 24% of patients in the low-dose MSC group and 6% of patients in the high-dose MSC group ($P = .022$), whereas no patients from control met this threshold. Furthermore, patients with underlying osteoarthritis who received the MSCs experienced a significant improvement in their symptoms (measured by VAS pain scores) compared with control ($P < .001$).

SUMMARY

The use of biologic augmentation, such as PRP and stem cell therapy, has great potential because it may help restore a tissue's form and function. Early clinical data demonstrate that the use of PRP and autologous stem cells for musculoskeletal treatment is safe; however, significant evidence of efficacy has yet to be shown. Certain biologic preparations appear to be more effective for certain indications, but the nuances of applying each therapy are still evolving. Interpreting existing literature has also been complicated because of a lack of standardization of the PRP or specific cell populations. Regulatory issues governing the use of stem cells and orthobiologics have continued to be a barrier in the development of these products. Despite these limitations, this technology has shown promise and will likely become an important component of the clinical treatment armamentarium in orthopedic and musculoskeletal medicine.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Mshra AK, Skrepnik NV, Edwards SG, et al. Efficacy of platelet-rich plasma for chronic tennis elbow: a double-blind, prospective, multicenter, randomized controlled trial of 230 patients. *Am J Sports Med*. 2014;42(2):463–471.

Level of Evidence:

II, randomized controlled study

Summary:

Leukocyte-rich PRP (LR-PRP) was evaluated against local injection of bupivacaine in 230 patients with lateral epicondylitis refractory to conservative therapy for at least 3 months. LR-PRP significantly improved symptoms at 24 weeks follow-up compared with active control.

Citation:

Jo CH, Shin JS, Shin WH, et al. Platelet-rich plasma for arthroscopic repair of medium to large rotator cuff tears: a randomized controlled trial. *Am J Sports Med*. 2015;43(9):2102–2110.

Level of Evidence:

I, randomized controlled study

Summary:

Seventy-four patients underwent arthroscopic repair of medium to large rotator cuff tears and were randomized to PRP-augmented repair or conventional repair. At 3 months follow-up, there was no significant in clinical symptom or function scores. However, at 1 year postoperatively, patients treated with PRP had a significantly decreased retear rate.

Citation:

Dragoo JL, Wasterlain AS, Braun HJ, et al. Platelet-rich plasma as a treatment for patellar tendinopathy: a double-blind, randomized controlled trial. *Am J Sports Med*. 2014;42(3):610–618.

Level of Evidence:

I, randomized controlled trial

Summary:

This randomized controlled trial evaluated dry needling with injection of LR-PRP for the treatment of patellar tendinopathy versus dry needling only. All patients were given standard eccentric exercises. Injection of LR-PRP significantly accelerated recovery from patellar tendinopathy. The benefit of PRP dissipated over time.

Citation:

Michalek J, Moster R, Lukac L, et al. Autologous adipose tissue-derived stromal vascular fraction cells application in patients with osteoarthritis. *Cell Transplant*. 2015.

Level of Evidence:

III, multicenter case controlled study

Summary:

Multicenter case control study involving 1114 patients with knee or hip osteoarthritis were injected with adipose tissue-derived stem cells (ADSCs) plus PRP and followed up for a median of 17.2 months. At 12 months of follow-up, 91% of patients had at least 50% improvement in symptoms and 63% of patients had at least 75% symptom improvement; 0.9% of patients were nonresponders.

Citation:

Patel S, Dhillon MS, Aggarwal S, et al. Treatment with platelet-rich plasma is more effective than placebo for knee osteoarthritis: a prospective, double-blind, randomized trial. *Am J Sports Med.* 2013;41:356–364.

Level of Evidence:

I, randomized controlled trial

Summary:

A total of 78 patients (156 knees) with bilateral osteoarthritis were divided randomly into three groups: group A received a single

injection of PRP, group B received two injections of PRP 3 weeks apart, and group C received a single injection of normal saline solution. A single dose of white blood cell–filtered PRP (LP-PRP) in concentrations of 10 times the normal amount was as effective as two serial injections to alleviate symptoms in patients with the early presentation of knee osteoarthritis. However, the results deteriorated after 6 months.

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